ELEC 231– MATHEMATICAL METHODS I FOR ELECTRICAL AND COMPUTER ENGINEERING

Course Syllabus – Fall 2026

This is your course syllabus. Please download the file and keep it for future reference.

LAND ACKNOWLEDGEMENT

Queen's University is situated on traditional Anishinaabe and Haudenosaunee Territory. See: http://www.queensu.ca/encyclopedia/t/traditional-territories

INCLUSIVITY STATEMENT

Queen's students, faculty, and staff come from every imaginable background – small towns and suburbs, urban high rises, Indigenous communities, and from more than 100 countries around the world. You belong here: https://www.queensu.ca/inclusive/.

TEACHING TEAM

Course Instructor

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Teaching Assistants

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ELEC 231 (F 3-0-0.5 3.5)

Course Description

Complex numbers, first/second-order ordinary differential equations, analytical methods for solving differential equations including the Laplace transform, convolution, linear systems of differential equations, numerical methods for solving differential equations, mathematical models for physical/natural phenomena and those arising in electrical and computer engineering applications. Exclusions: MTHE 225

(21/11/0/10/0) (Mathematics/Natural Sciences/Complementary Studies/Engineering Science/Engineering Design)

Pre-Requisite Knowledge

APSC 171 (Calculus I), APSC 172 (Calculus II), APSC 174 (Introduction to Linear Algebra)

Course Learning Outcomes (CLO)

By the end of this course, students should be able to:

CLO	Description	Indicator
CLO 1	Compute the general or particular solutions to linear and nonlinear differential equations, and systems of differential equations to model practical applications in electrical and computer engineering.	KB-Mathematics (Computational proficiency)
CLO 2	Understand the implications of the existence and uniqueness theorem and linear independence/dependence properties of solutions to differential equations.	KB-Mathematics (Mathematical reasoning)
CLO 3	Develop mathematical models describing physical/natural phenomena, with emphasis on problem based learning using authentic real-world electrical and computer engineering applications.	KB-NS
CLO 4	Use the method of Laplace transform to solve differential equations with applications in communication, signal processing, control systems	KB-ES
CLO 5	Model an RLC (resistor, inductor, capacitor) circuit using ordinary differential equations.	KB-ES
CLO 6	Numerically solve differential equations to provide tractable solutions for complex engineering systems.	KB-Mathematics

WEEKLY COURSE LEARNING OUTCOMES

Week	Learning Outcomes	Assessment
1	**General Introduction** By the end of this week, learners will be able to: • Mathematically model using differential equations • Understand the algebra of complex numbers • Represent complex numbers with exponentials, point, vector and polar forms	
2	**Initial value problems** By the end of this week, learners will be able to: • Classify different types of differential equations • Define a solution to a 1st order ODE and Initial Value Problem (IVP) • Use techniques for solving special 1st order ODEs in 1D	
3	**Techniques for first order ODEs; basic theory** By the end of this week, learners will be able to: • Use techniques for solving special 1st order ODEs in 1D: Separable and Linear • Apply existence and uniqueness of solutions to 1st order ODEs in 1D	
4	**Exact first order ODEs; introduction to second order ODEs** By the end of this week, learners will be able to: • Solve exact ODEs and use integrating factors to make an inexact ODE exact • Solve homogeneous second order ODEs with constant coefficients	

Week	Learning Outcomes	Assessment
5	**Homogeneous second order ODEs with constant coefficients** By the end of this week, learners will be able to: • Solve homogeneous second order ODEs with constant coefficients • Apply methods to RLC circuit with constant impressed voltage • Discuss basic existence and uniqueness of solutions to linear 2nd order ODEs in 1D	
6	**Theory of second order ODEs; nonhomogeneous systems** By the end of this week, learners will be able to: • Apply theorems on existence and uniqueness of solutions to linear 2nd order ODEs in 1D • Use the method of reduction of order • Solve nonhomogeneous equations using the method of undetermined coefficients	
7	**Laplace transforms** By the end of this week, learners will be able to: • Define the Laplace transform • Derive the Laplace transform of basic functions and properties of the Laplace transform Use the Heaviside step function and its Laplace transform	Midterm Test [CLO1, CLO2, CLO3, CLO5]
8	**Using Laplace transforms to solve initial value problems** By the end of this week, learners will be able to: • Use the Laplace transform to solve initial value problems • Solve ODEs with discontinuous forcing functions • Solve ODEs with Impulse functions	

Week	Learning Outcomes	Assessment
9	**Convolution theorem and introduction to systems of first order equations** By the end of this week, learners will be able to: • Use the convolution theorem • Solve systems of first order equations Apply the basic theory of first order linear systems	
10	**Linear systems** By the end of this week, learners will be able to: • Solve homogeneous linear systems with constant coefficients with real eigenvalues • Complex conjugate eigenvalues	
11	**Linear systems and phase-plane diagrams** By the end of this week, learners will be able to: • Solve homogeneous linear systems with constant coefficients with repeated eigenvalues • Solve nonhomogeneous linear systems and variation of parameters • Draw the phase-plane diagram associated with linear systems • Discuss the stability of equilibria of simple autonomous systems	

Week	Learning Outcomes	Assessment
12	**Numerical solutions of differential equations** Euler methods and error analysis Runge-Kutta methods Apply numerical solution methods to real-world industrial problems.	
Exam		**Final Exam [CLO1, CLO2, CLO3, CLO4, CLO5, CLO6]**

COURSE EVALUATION

Assessment Weighting

Assessment Tool	Due Date (before 23:59 ET)	Weight	Alignment with CLOs
Assignments 1-10	10 non-holiday Mondays	**20%**	**1,2,3,4,5,6**
Midterm test	Friday evening, October 30th (2hrs)	**30%**	**1,2,3,5**
Final Exam	**Please see the Final Exam schedule posted by OUR Exams Office**	**50%**	**1,2,3,4,5,6**
		100%	

ASSESSMENT DESCRIPTIONS

There are ten assignments in this course. They will be posted in onQ the Friday before the Monday and Wednesday Tutorials. Each assignment will require you to submit your solution in onQ **before** the tutorial, during which the solution will be discussed. Each time you will receive 2 marks for a legitimately-attempted solution regardless of its correctness, and 0 otherwise.

Midterm Week 7 (30 marks)
[CLO1, CLO2, CLO3, CLO5]

Final Exam

The final exam is closed book. Students must write their exam on the day and time scheduled by the University. You should not schedule vacations, travel, etc. during the exam period. The Term and Session Dates will indicate the final exam period session dates in each term.

GRADING

All assessments in this course will receive numerical percentage marks. The final grade you receive for the course will be derived by converting your numerical course average to a letter grade according to the established Grade Point Index.

Feedback on Assessments

The teaching team will provide feedback on graded activities. You can expect feedback on your assessments within 14 days of the due date.

Accessing Your Final Grade

Your final grades will show on SOLUS. Official transcripts showing final grades will be available on the Official Grade Release Date. Please note that in official transcripts, a mark of IN (incomplete) is considered a grade, and your transcript is released with this grade.

COURSE MATERIALS

Required Textbook

Nilsson and Reidel, Electric Circuits, 12th Edition (Electronic)

ISBN: 978-0137648375

E-textbook is available at: https://mylabmastering.pearson.com/?courseId=12900656
5 options from $77.99

This text is also used for ELEC 221, so there is zero additional textbook cost for ELEC 231.

Editions 10 and 11 are also fine.

All other course material is accessible via OnQ.

Required Calculator

A Casio 991 is required. **ONLY** this type of non-programmable, non-communicating calculator will be allowed during tests and exams.

Suggested Time Commitment

This course represents a study period of one semester spanning 12 weeks. Learners can expect to invest on average 7-9 hours per week in this course. Learners who adhere to a pre-determined study schedule are more likely to successfully complete the course.

COURSE COMMUNICATION

QUESTIONS ABOUT COURSE MATERIAL

Questions or comments regarding the course material that can be of benefit to other students should be posted in the Q&A forum on the class website. The instructor, TAs, and students are encouraged to answer these questions directly in the discussion forum for the benefit of everyone in the course.

COURSE ANNOUNCEMENTS

The instructor will routinely post course news in the Announcements section on the main course homepage on OnQ. Please sign up to be automatically notified by email when the instructor posts new information in the Announcements section. Instructions on how to modify your notifications are found in the **Begin Here** section of the onQ course site.

OFFICE HOURS

In addition to interaction in the Q&A discussion forums, you will have the opportunity to interact with either a TA or the instructor during office hours. The instructor will provide a schedule of availability at the beginning of the term.

CONFIDENTIAL MATTERS

If you have a confidential matter you would like to discuss with your instructor, their contact details are on the first page of this document. Expect email replies within 48 hours, not counting weekends and holidays.

ABSENCES (ACADEMIC CONSIDERATIONS) AND MISSED ASSIGNMENTS

For information on academic considerations due to extenuating circumstances, please review the information on the Smith Engineering website. Note that unacceptable reasons include extra-curricular activities, travel plans, generally behind on schoolwork, etc. Do not schedule travel during midterms and final exams, as travel is not an acceptable reason for granting academic considerations.

In cases of an **approved** consideration request (i.e., if a quiz/midterm is missed for an acceptable reason), the marks will be redistributed to the rest of the individually written components of the course, if an assignment is missed, the marks will be redistributed to the rest of the assignments.

Late Policy

In the event of extenuating circumstances, you must follow the policies for requesting an academic consideration (as described above). In the absence of an approved consideration request, the normal late penalty will apply as described in the assignment or any course/departmental policies.

STANDARD QUEEN'S AND SMITH ENGINEERING POLICIES

Netiquette

In this course, you may be expected to communicate with your peers and the teaching team through electronic communication. You are expected to use the utmost respect in your dealings with your colleagues or when participating in activities, discussions, and online communication.

Following is a list of netiquette guidelines. Please read them carefully and use them to guide your online communication in this course and beyond.

1. Make a personal commitment to learn about, understand, and support your peers.
2. Assume the best of others and expect the best of them.
3. Acknowledge the impact of oppression on the lives of other people and make sure your writing is respectful and inclusive.
4. Recognize and value the experiences, abilities, and knowledge each person brings.
5. Pay close attention to what your peers write before you respond. Think through and re-read your writings before you post or send them to others.
6. It's alright to disagree with ideas, but do not make personal attacks.
7. Be open to be challenged or confronted on your ideas and challenge others with the intent of facilitating growth. Do not demean or embarrass others.
8. Encourage others to develop and share their ideas.

Student Code of Conduct

Queen's University values maintaining an environment free of, and will not tolerate, harassment, discrimination, and reprisal. The Student Code of Conduct applies to all students at Queen's. It outlines the activities and behaviours that could be considered Non-Academic Misconduct (NAM). The Code also describes the NAM process and the sanctions that could be imposed on a student found responsible for a violation. All students should be familiar with the Student code of conduct and related policies on sexual violence prevention and response and harassment and discrimination prevention and response.
https://www.queensu.ca/nonacademicmisconduct/policies

COPYRIGHT

Course materials created by the course instructor, including all slides, presentations, synchronous and asynchronous course recordings, handouts, tests, exams, and other similar course materials, are the intellectual property of the instructor. It is a departure from academic integrity to distribute, publicly post, sell or otherwise disseminate an instructor's course materials or to provide an instructor's course materials to anyone else for distribution, posting, sale or other means of dissemination, without the instructor's ***express consent***. A student who engages in such conduct may be subject to penalty for a departure from academic integrity and may also face adverse legal consequences for infringement of intellectual property rights and, with respect to recordings, potentially privacy violations of other students.

ACADEMIC INTEGRITY

As an engineering student, you have made a decision to join us in the profession of engineering, a long-respected profession with high standards of behaviour. As future engineers, we expect you to behave with integrity at all times. Please note that Engineers have a duty to:

- Act at all times with devotion to the high ideals of personal honour and professional integrity.
- Give proper credit for engineering work

The standard of behaviour expected of professional engineers is explained in the Professional Engineers Ontario Code of Ethics. Information on policies concerning academic integrity is available in the Queen's University Code of Conduct, in the Senate Academic Integrity Policy Statement, on the Smith Engineering website, and from your instructor.

Departures from academic integrity include plagiarism, use of unauthorized materials or services, facilitation, forgery, falsification, unauthorized use of intellectual property, and collaboration, and are antithetical to the development of an academic community at Queen's. Given the seriousness of these matters, actions which contravene the regulation on academic integrity carry sanctions that can range from a warning or the loss of grades on an assignment to the failure of a course to a requirement to withdraw from the University.

In the case of online or remotely proctored exams, impersonating another student, copying from another student, making information available to another student about the exam questions or possible answers, posting materials to online services, communicating with another person during an exam or about an exam during the exam window, or accessing unauthorized materials, including internet sources and using unauthorized materials, including smart devices, are actions in contravention of academic integrity.

USE OF GENERATIVE ARTIFICIAL INTELLIGENCE (GENAI) TOOLS:

Students must submit their own work and cite the work that is not theirs. Any other use constitutes a Departure from Academic Integrity.

GenAI tools are NOT permitted in this course: Use of GenAI tools are not allowed in any part of student work for this course. Submitting AI-generated content constitutes a Departure from Academic Integrity as defined by University Academic Integrity procedures.

Invalid Exams

An exam may be declared invalid in case of an interruption in an in-person examination; if the instructions in a remote or online exam were not followed; if the student uploads wrong materials; or if a situation arises where the integrity of the exam cannot be verified. If an exam is declared invalid, the student may be granted a re-write.

ACADEMIC AND STUDENT SUPPORT

Queen's has a robust set of supports available to you including the Library, Student Academic Success Services (Learning Strategies and Writing Centre), and Career Services. Learners are encouraged to visit the Smith Engineering Current Students web portal for information about various other policies such as academic advisors, registration, student exchanges, awards and scholarships, etc. Students are also encouraged to review the information that is available in the EngQ Hub, posted in onQ.

Absences (Academic Considerations) and Academic Accommodations

For academic accommodations and considerations please review the information on the Smith Engineering website.

Accommodations for Disabilities

Queen's University is committed to working with students with disabilities to remove barriers to their academic goals. Queen's Student Accessibility Services (QSAS), students with disabilities, instructors, and faculty staff work together to provide and implement academic accommodations designed to allow students with disabilities equitable access to all course material (including in-class as well as exams). If you are a student currently experiencing barriers to your academics due to disability related reasons, and you would like to understand whether academic accommodations could support the removal of those barriers, please visit the QSAS website (https://www.queensu.ca/studentwellness/accessibility-services) to learn more about academic accommodations. To start the registration process with QSAS, click the ***Access Ventus*** button found on the Ventus student portal: https://www.queensu.ca/studentwellness/accessibility-services/ventus

Ventus is an online portal that connects students, instructors, Queen's Student Accessibility Services, the Exam's Office, and other support services in the process to request, assess, and implement academic accommodations. To learn more about Ventus, visit A Visual Guide to Ventus for Students:
https://www.queensu.ca/ventus-support/students/visual-guide-ventus-students

For questions or assistance with requesting Academic Consideration or Accommodation, contact the Smith Engineering Program Advisor (Accommodations and Considerations) at engineering.aac@queensu.ca

Every effort has been made to provide course materials that are accessible. For further information on accessibility compliance of the educational technologies used in this course, please consult the links below.

EDUCATIONAL TECHNOLOGY	ACCESSIBILITY COMPLIANCE INFORMATION
onQ (Brightspace Learning Management System by D2L)	https://www.d2l.com/accessibility/standards/
MS-Teams	https://support.microsoft.com/en-us/office/accessibility-support-for-microsoft-teams-d12ee53f-d15f-445e-be8d-f0ba2c5ee68f
om	https://zoom.us/accessibility

If you find any element of this course difficult to access, please discuss with your instructor how you can obtain an accommodation.

REQUESTS FOR RELIGIOUS AND OTHER HUMAN RIGHTS-BASED ACCOMMODATIONS IN SMITH ENGINEERING

RELIGIOUS ACCOMMODATIONS

Students in need of accommodation for religious observance are asked to complete the Religious Observance and Academics activities form within a week of receiving their course syllabus (each term) and submit the form via email to the Engineering Program Advisor, (Accommodations & Considerations), engineering.aac@queensu.ca

- Please use this form: Religious Observance and Academics Activities
- Alternative assignments are considered a "reasonable accommodation" under the Ontario Human Rights Code
- Please refer to the Queen's Multi-faith Calendar for all of this year's faith dates
- Students with questions about their rights and responsibilities regarding religious accommodations can contact chaplain@queensu.ca

Other human-rights based Accommodations

Engineering students who need accommodations on Human Rights grounds are asked to contact, in confidence, the Engineering Program Advisor, (Accommodations & Considerations) engineering.aac@queensu.ca or to call 613-533-6000 x 78013.

Technical Support

Some basic comfort level with basic hardware and software skills are required for this course. If you require technical assistance, please contact Technical Support.

Supportive Personal Counselling

If at any time you find yourself feeling overwhelmed, anxious, sad, lonely, or distressed, consider confidential personal counselling and wellness services offered by Smith Engineering and the Queen's student wellness services.